Introduction

This coursework was done by Kamile Kavaliauskaite H00460022. It analyses a monthly water consumption for 857 households. It follows Gamma distribution. The given data is right-skewed. This work includes two histograms done on R, and some calculations and formula derivations done with pen on paper. The calculations done for MME, MLE, and confidence intervals, also a simulation of future consumption values.

Question 1

The sample mean:	$\bar{x} \approx 13.9477$ m³
Sample Standard Deviation:	s ≈ 6.9645 m³
Sample median:	median = 12.8177 m³
Quantiles:	25%: 9 m³
	75%: 17.7 m³

The histogram for this data is right-skewed, which shows that the majority of households consume between 7m³ to 21m³, and only a decreasingly small amount of households consume more than 21m³. About a quarter of the households consume less than the majority, and a quarter of households consume more than 17.4 m³.

See the histogram below.

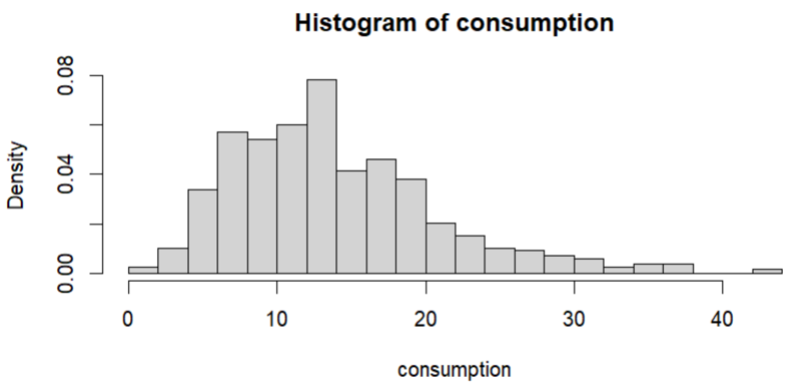
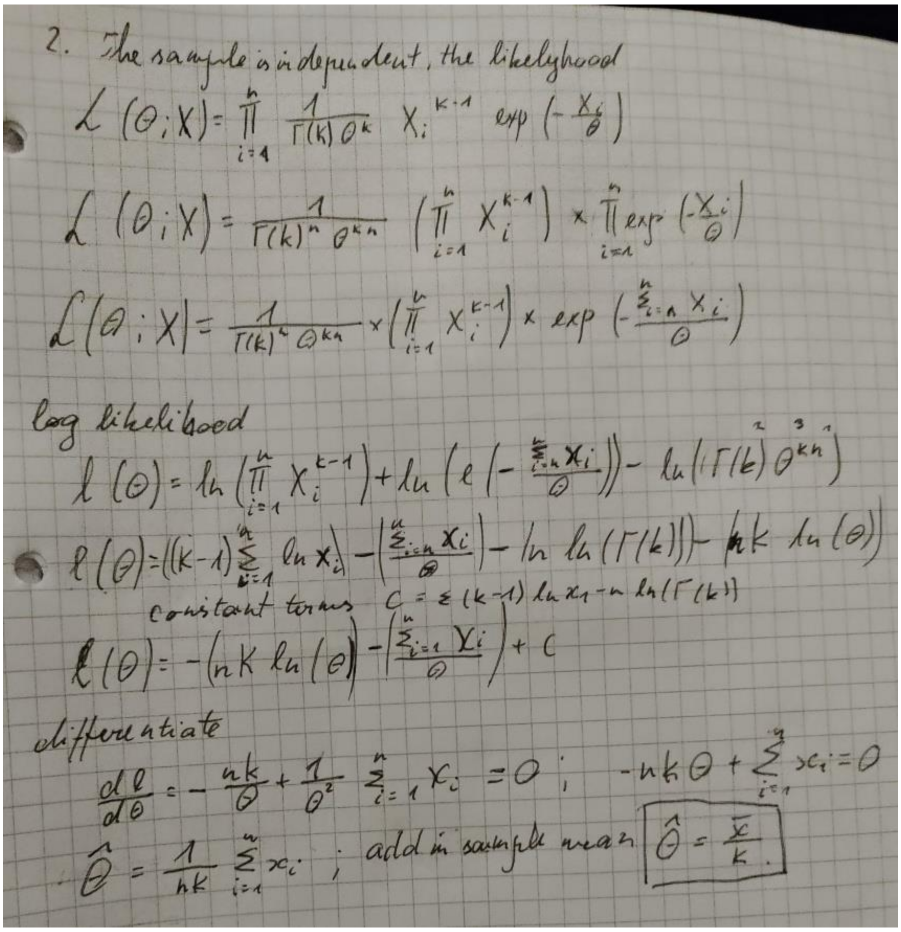


Figure 1: Histogram

Question 2

Here is the working of deriving the formula for the maximum likelihood estimator (MLE) for θ bellow.



2. The sample is independent, the likelyhood

$$\mathcal{L}(\theta;X) = \prod_{i=1}^{n} \frac{1}{\Gamma(k)\theta^k} X_i^{k-1} \exp\left(-\frac{X_i}{\theta}\right)$$

$$\mathcal{L}(\theta;X) = \frac{1}{\Gamma(k)^n \theta^{kn}} \left(\prod_{i=1}^{n} X_i^{k-1}\right) \times \prod_{i=1}^{n} \exp\left(-\frac{X_i}{\theta}\right)$$

$$\mathcal{L}(\theta;X) = \frac{1}{\Gamma(k)^n \theta^{kn}} \times \left(\prod_{i=1}^{n} X_i^{k-1}\right) \times \exp\left(-\frac{\sum_{i=1}^{n} X_i}{\theta}\right)$$

log likelihood

$$l(\theta) = \ln\left(\prod_{i=1}^{n} X_i^{k-1}\right) + \ln\left(e\left(-\frac{\sum_{i=1}^{n} X_i}{\theta}\right)\right) - \ln\left(\Gamma(k)\theta^{kn}\right)$$

$$\ell(\theta) = \left((k-1)\sum_{i=1}^{n} \ln x_i\right) - \left(\frac{\sum_{i=1}^{n} x_i}{\theta}\right) - \left(n \ln(\Gamma(k))\right) - nk \ln(\theta)$$

constant terms $C = \Sigma(k-1)\ln x_1 - n\ln(\Gamma(k))$

$$\ell(\theta) = -(nk \ln(\theta)) - \left(\frac{\sum_{i=1}^{n} X_i}{\theta}\right) + C$$

differentiate

$$\frac{d\ell}{d\theta} = -\frac{nk}{\theta} + \frac{1}{\theta^2}\sum_{i=1}^{n} x_i = 0 \; ; \quad -nk\theta + \sum_{i=1}^{n} x_i = 0$$

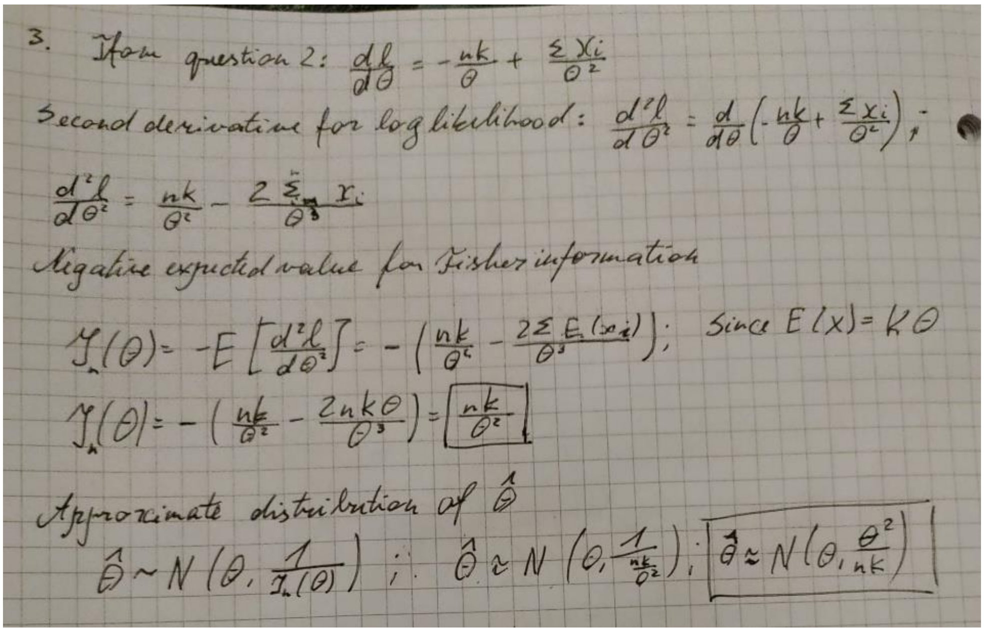
$$\hat{\theta} = \frac{1}{nk}\sum_{i=1}^{n} x_i \; ; \text{ add in sample mean } \boxed{\hat{\theta} = \frac{\bar{x}}{k}}$$

Figure 2: Question 2 (MLE)

The answer is $\hat{\theta} = \frac{\bar{x}}{\mathrm{k}}$

Question 3

Here is working for Fisher information for θ shown bellow.


3. From question 2: $\frac{dl}{d\theta} = -\frac{nk}{\theta} + \frac{\sum x_i}{\theta^2}$

Second derivative for log likelihood: $\frac{d^2 l}{d\theta^2} = \frac{d}{d\theta}\left(-\frac{nk}{\theta} + \frac{\sum x_i}{\theta^2}\right)$;

$\frac{d^2 l}{d\theta^2} = \frac{nk}{\theta^2} - \frac{2\sum_{i=1}^{n} x_i}{\theta^3}$

Negative expected value for Fisher information

$I_n(\theta) = -E\left[\frac{d^2 l}{d\theta^2}\right] = -\left(\frac{nk}{\theta^2} - \frac{2\sum E(x_i)}{\theta^3}\right)$; Since $E(x) = k\theta$

$I_n(\theta) = -\left(\frac{nk}{\theta^2} - \frac{2nk\theta}{\theta^3}\right) = \boxed{\frac{nk}{\theta^2}}$

Approximate distribution of $\hat{\theta}$

$\hat{\theta} \sim N\left(\theta, \frac{1}{I_n(\theta)}\right)$; $\hat{\theta} \approx N\left(\theta, \frac{1}{\frac{nk}{\theta^2}}\right)$; $\boxed{\hat{\theta} \approx N\left(\theta, \frac{\theta^2}{nk}\right)}$

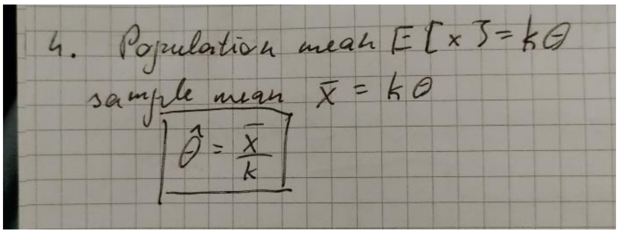
Figure 3: Fisher information and approximate distribution

The answer for Fisher information: I(θ) = $\frac{nk}{\theta^2}$

The answer for the approximate distribution: $\hat{\theta}$ ≈ N (θ, $\frac{\theta^2}{nk}$)

Question 4

MME for θ is shown below.


4. Population mean $E[x] = k\theta$

sample mean $\bar{x} = k\theta$

$\boxed{\hat{\theta} = \frac{\bar{x}}{k}}$

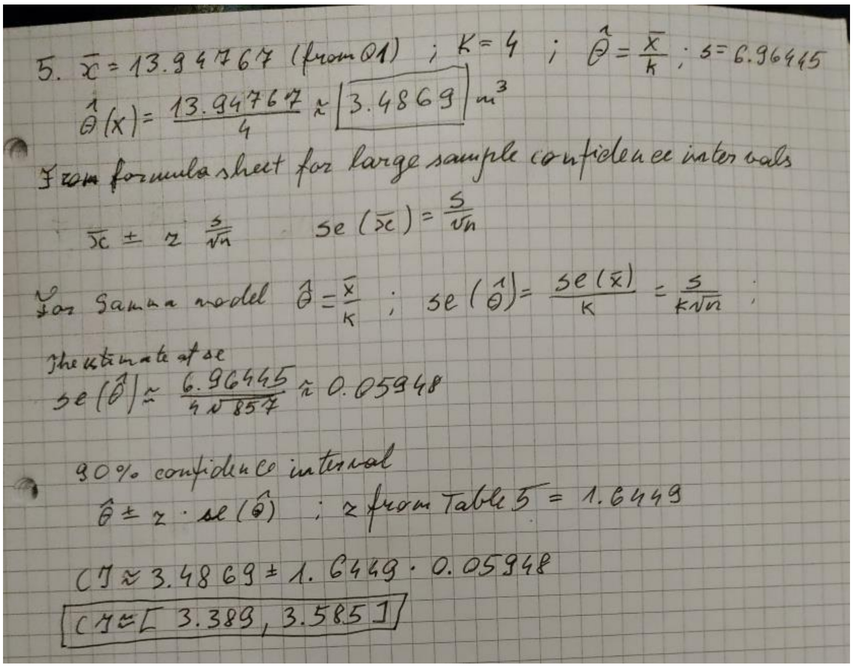
Figure 4:MME

The answer for MME: $\hat{\theta} = \frac{\bar{x}}{\mathrm{k}}$

Both MME and MLE are the same, because k is fixed and the Gamma mean depends only on θ.

Question 5

See the solution below.



5. $\bar{x} = 13.94767$ (from Q1) ; $K = 4$; $\hat{\theta} = \frac{\bar{x}}{k}$; $s = 6.96445$

$\hat{\theta}(x) = \frac{13.94767}{4} \approx \boxed{3.4869}\ m^3$

From formula sheet for large sample confidence intervals

$\bar{x} \pm z \frac{s}{\sqrt{n}}$ $se(\bar{x}) = \frac{s}{\sqrt{n}}$

For Gamma model $\hat{\theta} = \frac{\bar{x}}{k}$; $se(\hat{\theta}) = \frac{se(\bar{x})}{K} = \frac{s}{K\sqrt{n}}$;

The estimate of se

$se(\hat{\theta}) \approx \frac{6.96445}{4\sqrt{857}} \approx 0.05948$

90% confidence interval

$\hat{\theta} \pm z \cdot se(\hat{\theta})$; z from Table 5 = 1.6449

$CI \approx 3.4869 \pm 1.6449 \cdot 0.05948$

$\boxed{CI \approx [\,3.389, 3.585\,]}$

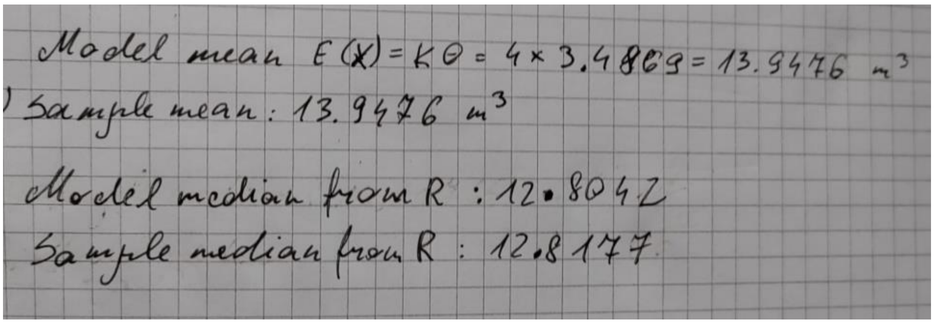
Figure 5: confidence interval

The $\hat{\theta}$ $(\bar{x})$ = 3.4869

Approximate 90% equal-tailed confidence interval for θ: [3.389, 3.585]

Question 6

Model mean and median vs sample mean and median comparison given below.



Model mean $E(X) = K\theta = 4 \times 3.4869 = 13.9476\ m^3$

Sample mean: $13.9476\ m^3$

Model median from R : 12.8042

Sample median from R : 12.8177

Figure 6: Sample vs Model (mean/median)

Model mean: 13.9476 Sample mean: 13.9477

Model median: 12.8042 Sample median: 12.8177

The gamma model reproduces the mean accurately. The mean and median is almost the same, with a slight variation. The model fits the data very well.

Question 7

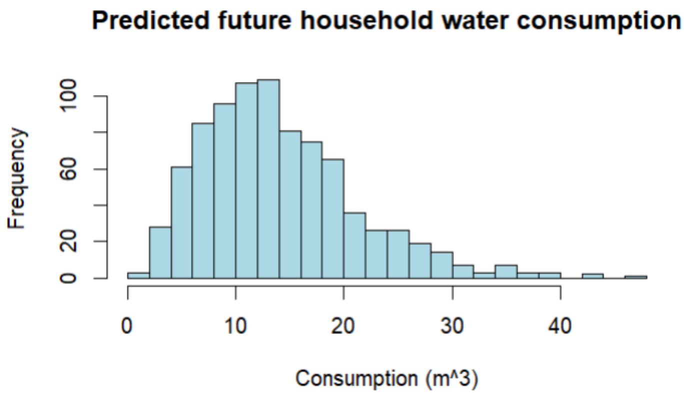


Figure 7: Future consumption prediction histogram

q25	Named num 9.27
q5	Named num 4.65
q75	Named num 17.2
q95	Named num 25.4

Figure 8: 95%, 75%, 25%, 5% quantiles

pred_mean	13.8163718712559
pred_median	12.9120655347229
pred_sd	6.48725318175351

Figure 9: Mean, median, standard deviation

Conclusions

It is clearly visible that when plotted on a histogram, the data is right-skewed. Most households consume between 7 m³ to 21 m³ per month. The mean of the sample is 13.95 m³, and the median is 12.3 m³. The standard deviation shows that consumption varies 6.96 m³ from the average consumption. The θ scale parameter and MME are approximately 3.49. The 90% confidence interval is approximately [3.37, 3.6]. The model mean matches the sample mean exactly, and the model median 12.8 is slightly lower than the sample 12.82, so the model fits the sample well. The future consumption mean is approximately 13.82, and the median 12.9, which is a very close range. This shows that the model can make realistic predictions.

References

Heriot-Watt university (2025) *F20SA / F21SA Formula sheet.* [PDF] Available at: Canvas (Accessed: 21/11/2025)

Heriot-Watt university (2025) *F20SA / F21SA Statistical Modelling and analysis.* [PDF] Available at: Canvas (Accessed: 21/11/2025)

Appendix

```
# 1. Histogram and numerical summaries calculation

# Load data from hwcData.csv
cwdata <- read.csv("hwcData.csv", header = FALSE)

# Consumption values
consumption <- cwdata$V1

# Number of elements
n <- length(consumption)
k <- 4

# Mean
mean_con <- mean(consumption)
# Standard deviation
stan_dev <- sd(consumption)
# Median
median_con <- median(consumption)
# Quantiles

q25 <- quantile(consumption, 0.25)
q75 <- quantile(consumption, 0.75)

# Histogram
hist(consumption, breaks = 30, prob = TRUE)

# 2. MLE - maximum likelihood estimator
mle_con <- mean_con / k
```

```
con_vals <- seq(min(consumption), max(consumption))

lines(con_vals, dgamma(con_vals, shape = k, scale = mle_con), col = "red", lwd = 2,
lty = 2)


# 6 model median

k <- 4

theta_hat <- 3.4869175


model_median <- qgamma(0.5, shape = k, scale = theta_hat)



# 7. predicted future consumptions and their mean

theta_hat <- mean_con / k

rate <- 1/theta_hat

rgamma(n, k, rate)


# 857 predict future consumptions

pred_cons <- rgamma(n, k, rate)


# Plot new histogram for predicted consumptions

hist(pred_cons, breaks = 30, col = "lightblue", main = "Predicted future household
water consumption", xlab = "Consumption (m^3)")


# predicted mean

pred_mean <- mean(pred_cons)

# pred sd

pred_sd <- sd(pred_cons)

# median pred

pred_median <- median(pred_cons)

# quantiles
```

```
pred_q25 <- quantile(pred_cons, 0.25)
pred_q75 <- quantile(pred_cons, 0.75)
_cons, 0.95)
```